# What Influences Listing Prices in Albany? An Airbnb Analysis

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## Abstract

Pricing in short-term rental markets reflects a combination of property features, host characteristics, and market competition. Understanding which factors are most strongly associated with listing prices can provide insight into how hosts market their properties and how guests evaluate accommodation options. This report analyzes 478 short-term rental listings in Albany using publicly available data compiled by Inside Airbnb from listings on Airbnb. Multiple linear regressions were used to examine relationships between listing price and a range of listing and host characteristics. We found listings that accommodate more guests and those hosted by superhosts were associated with higher prices, while listings categorized as private rooms or hotel rooms were priced lower than entire homes. Additionally, listings that allowed instant booking and those with a greater number of reviews were associated with lower prices. The findings suggest that both listing and host attributes contribute to price differences across short-term rental listings in Albany.

## Introduction

Online short-term rental platforms such as Airbnb have significantly changed the way people travel and book accommodations over the last two decades. For instance, instead of staying exclusively in traditional hotels, travelers can now rent private rooms, apartments, or entire homes from individual hosts. As the popularity of these platforms has grown, the variation in listing prices has grown as well. Knowing what drives these differences in price has become an important question for both hosts and guests participating in the short-term rental market.

In Albany, the New York state capital and a regional hub for tourism and government travel, Airbnb listings range from small private rooms to entire homes that accommodate larger groups. These listings differ in several key features, such as the number of bedrooms, bathrooms, beds, and the maximum number of guests they can accommodate. Host characteristics, such as whether the host is a "superhost," their acceptance rate, or whether the listing allows instant booking, could also influence pricing. Understanding the factors that influence Airbnb prices allows hosts to price their listings competitively and guests to better assess their lodging options. Through multiple linear regressions, this report aims to determine which listing and host characteristics are most strongly associated with Airbnb prices in Albany.

## Methods

The data used in this analysis were obtained from Inside Airbnb, which compiles publicly available information from listings on Airbnb. The dataset contains 478 observations, with each row representing a listing in Albany and each column representing a listing or host characteristic.

Multiple linear regressions were used to examine the relationship between listing characteristics and nightly price. We used the natural logarithm of nightly price as our dependent variable to better satisfy linear regression assumptions. The predictors considered were: the number of reviews, the number of listings a host had, maximum listing capacity, the number of bathrooms, bedrooms, and beds, the minimum number of nights to stay, the maximum number of nights to stay, the host's acceptance rate, and the host's rating, included as continuous variables; and the listing's instant booking capacity, the host's superhost status, and the listing's room type, included as categorical variables. Listing property type was excluded to decrease model complexity and improve coefficient estimate stability.

A full model including all predictors was first fit. Best subset selection was then used to identify a more parsimonious model. To assess the impact of high leverage points, observations with leverage greater than $2p/n$ for the number of parameters $p$ and observations $n$ were identified as high-leverage points and removed, and best-subsets regression was repeated. We report results with high leverage points removed to ensure that the model is not disproportionately influenced by a few extreme observations. Statistical significance was evaluated with the threshold $\alpha = 0.05$, and mean estimates were reported with 95% confidence intervals. After the final model was constructed, bootstrapping was used to assess if listings with missing prices were significantly different from listings with prices (for technical details, see Appendix B). The bootstrap statistic considered was the difference in means for a given predictor for listings that do not vs. do have observed prices.

## Results

Table 1 (see Appendix A) gives summary statistics on the listings for each variable of interest. The mean listing price was \$111 with a standard deviation of \$87, with prices following an approximately normal distribution. Some listings had very high prices, with the maximum price in the data being \$960. Out of the 478 observations, 360 listings were not instantly bookable (hence were not commercial listings) and 349 were the entire home/apartment.

The results of the multiple linear regression models are presented in Table 2 (see Appendix A). Column (1) gives coefficient estimates, confidence intervals, and $p$-values associated with a regression on all predictors, excluding property type. Columns (2) and (3) give these values for a regression model built using best subsets selection, including and excluding high leverage points, respectively. We ran our final analysis using the variables included in Column (3). We found several predictors consistently associated with listing price. Listings that accommodated more guests were associated with higher prices, with a 12.4% increase in price for each additional guest accommodated (95%CI: $(0.096, 0.152); p < 0.001$). Listings hosted by superhosts were associated with a 13.3% increase in price compared to non-superhosts (95%CI: $(0.055, 0.211); p = 0.001$).

Additionally, some predictors were associated with lower listing prices. Listings classified as private rooms or hotel rooms were associated with a 23.1% decrease in price relative to an entire home (95%CI: $(-0.326, -0.136); p < 0.001$). Also, listings that allowed instant booking were associated

with a 13.3% decrease in price (95%CI: $(-0.218, -0.048); p = 0.003$). A greater number of reviews was associated with 0.04% decrease in price per additional review (95%CI: $(-0.001, -0.0001); p = 0.019$). Similarly, an increase in the number of listings managed by a host was associated with a 0.6% decrease in price (95%CI: $(-0.010, -0.001); p = 0.025$).

We also found that a greater number of beds was associated with slightly higher prices ($\beta = 0.042; 95\%\text{CI}: (-0.007, 0.091); p = 0.093$), although this effect was not consistently statistically significant across models. The maximum nights allowed was associated with a small increase in price ($\beta = 0.0001; 95\%\text{CI}: (0.00001, 0.0002); p = 0.025$). Other predictors, including the number of bathrooms, number of bedrooms, minimum nights, and review scores, were not consistently associated with listing price across the models.

In the data, 42 listings do not include prices. The variables that were not significantly different for listings with prices that were vs. were not missing are the number of listings a host had (CI: $(-1.9932, 2.3003)$), accommodation capacity (CI: $(-0.7642, 0.5406)$), the number of bedrooms (CI: $(-0.1841, 0.4791)$), the minimum (CI: $(-0.53, 34.48)$) and maximum number of nights (CI: $(-208.3, 3.9)$), and the host's acceptance rate (CI: $(-19.292, 2.598)$). The variables that were significantly different were the number of reviews (CI: $(-60.29, -32.08)$), a host's instant booking status (CI: $(0.0436, 0.3521)$), a host's rating (CI: $(-0.6386, -0.0885)$), and their superhost status (CI: $(-0.4446, -0.1903)$).

## Discussion

We found that one of the strongest predictors of AirBnb listing price was accommodation capacity, with listings able to host more guests tending to have higher prices. This pattern is intuitive as larger properties offer more space and amenities and are suitable for groups, allowing hosts to charge higher nightly rates. Similarly, private rooms or hotel rooms were consistently priced lower than entire homes. Entire homes typically provide greater privacy and access to the full property, which likely justifies higher prices.

Host characteristics also appeared to influence pricing. Listings hosted by superhosts were associated with higher prices, which may reflect the greater reputation and trust associated with hosts who consistently receive high ratings and maintain strong hosting standards. Interestingly, listings that allowed instant booking were associated with slightly lower prices. One possible explanation is that hosts offering instant booking may be more likely to be corporate hosts, who use their scale to manage high-volume bookings more efficiently.

Surprisingly, several variables that were expected to influence price, such as the number of bedrooms, bathrooms, or minimum nights, were not consistently associated with price in the models. This may be because these characteristics are partially captured by other variables, such as accommodation capacity, or because there is relatively little variation in these features among listings in the dataset.

However, this analysis has certain limitations. First, the dataset includes only 478 listings, which may limit the ability to detect smaller effects and not fully represent the broader Airbnb market. Additionally, the data were obtained from publicly available listings and may not capture all factors that influence pricing, such as neighborhood location, amenities, seasonal demand, or host pricing strategies. Finally, the presence of observations with missing prices may have biased our estimates' magnitudes, directions, and statistical significance levels.

## Conclusion

This report examines how listing and host characteristics are associated with Airbnb prices in Albany. We found that accommodation capacity, room type, and superhost status were among the strongest predictors of price. In particular, listings that accommodated more guests and those hosted by superhosts tended to have higher prices, while private rooms or hotel rooms were typically priced lower than entire homes. Overall, the analysis provides insight into factors associated with Airbnb pricing in Albany and may help both hosts and guests better understand the characteristics that influence variation in listing prices.

# Appendix

## Appendix A

Table 1: Summary Statistics on Albany Airbnb Listings (N = 478)

| Variable | N | Mean | SD | Minimum | 25th Pctl. | 75th Pctl. | Maximum |
|---|---|---|---|---|---|---|---|
| Price | 436 | 111 | 87 | 24 | 68 | 121 | 960 |
| Number of Reviews | 478 | 59 | 111 | 0 | 3 | 67 | 964 |
| Instant Booking? | 478 | | | | | | |
| ... FALSE | 360 | 75% | | | | | |
| ... TRUE | 118 | 25% | | | | | |
| Calculated Host Listing Count | 478 | 7.5 | 7.1 | 1 | 2 | 13 | 23 |
| Property Type | 478 | | | | | | |
| ... Entire condo | 7 | 1% | | | | | |
| ... Entire cottage | 1 | 0% | | | | | |
| ... Entire guest suite | 3 | 1% | | | | | |
| ... Entire guesthouse | 1 | 0% | | | | | |
| ... Entire home | 74 | 15% | | | | | |
| ... Entire loft | 2 | 0% | | | | | |
| ... Entire place | 2 | 0% | | | | | |
| ... Entire rental unit | 245 | 51% | | | | | |
| ... Entire townhouse | 10 | 2% | | | | | |
| ... Entire vacation home | 3 | 1% | | | | | |
| ... Private room in bed and breakfast | 5 | 1% | | | | | |
| ... Private room in condo | 1 | 0% | | | | | |
| ... Private room in guest suite | 3 | 1% | | | | | |
| ... Private room in guesthouse | 1 | 0% | | | | | |
| ... Private room in home | 70 | 15% | | | | | |
| ... Private room in rental unit | 27 | 6% | | | | | |
| ... Private room in townhouse | 2 | 0% | | | | | |
| ... Private room in villa | 1 | 0% | | | | | |
| ... Room in aparthotel | 1 | 0% | | | | | |
| ... Room in hotel | 19 | 4% | | | | | |
| Room Type | 478 | | | | | | |
| ... Entire home/apt | 349 | 73% | | | | | |
| ... Hotel room | 1 | 0% | | | | | |
| ... Private room | 128 | 27% | | | | | |
| Accommodation Capacity | 478 | 3.4 | 2.4 | 1 | 2 | 4 | 16 |
| Number of Bathrooms | 437 | 1.2 | 0.64 | 0 | 1 | 1 | 7 |
| Number of Bedrooms | 473 | 1.6 | 1.1 | 0 | 1 | 2 | 9 |
| Number of Beds | 437 | 1.9 | 1.3 | 0 | 1 | 2 | 10 |
| Minimum Nights | 478 | 6.8 | 22 | 1 | 1 | 3 | 365 |
| Maximum Nights | 478 | 481 | 392 | 1 | 225 | 730 | 1125 |
| Host Acceptance Rate | 460 | 89 | 22 | 0 | 89 | 100 | 100 |
| Review Scores Rating | 410 | 4.7 | 0.42 | 1 | 4.7 | 5 | 5 |
| Host is Superhost | 472 | | | | | | |
| ... FALSE | 245 | 52% | | | | | |
| ... TRUE | 227 | 48% | | | | | |

Table 2: Multiple Linear Regression Results on Listing Price

| | *Dependent variable:* | | |
|---|---|---|---|
| | Log Price | | |
| | (1) | (2) | (3) |
| Intercept | 3.649*** | 3.691*** | 3.816*** |
| | (3.079, 4.220) | (3.204, 4.179) | (3.312, 4.320) |
| | p = 0.000 | p = 0.000 | p = 0.000 |
| Number of Reviews | −0.0004** | −0.0004*** | −0.0004** |
| | (−0.001, −0.0001) | (−0.001, −0.0001) | (−0.001, −0.0001) |
| | p = 0.013 | p = 0.004 | p = 0.019 |
| Instant Bookability (Reference = False) | −0.116*** | −0.137*** | −0.133*** |
| | (−0.202, −0.030) | (−0.220, −0.054) | (−0.218, −0.048) |
| | p = 0.009 | p = 0.002 | p = 0.003 |
| Calculated Host Listings Count | −0.004* | −0.005* | −0.006** |
| | (−0.009, 0.001) | (−0.009, 0.0001) | (−0.010, −0.001) |
| | p = 0.100 | p = 0.054 | p = 0.025 |
| Room Type (Reference = Entire Home) | −0.260*** | −0.253*** | −0.231*** |
| | (−0.356, −0.164) | (−0.341, −0.165) | (−0.326, −0.136) |
| | p = 0.00000 | p = 0.00000 | p = 0.00001 |
| Accommodation Capacity | 0.111*** | 0.117*** | 0.124*** |
| | (0.079, 0.142) | (0.091, 0.144) | (0.096, 0.152) |
| | p = 0.000 | p = 0.000 | p = 0.000 |
| Number of Bathrooms | −0.012 | | |
| | (−0.076, 0.052) | | |
| | p = 0.712 | | |
| Number of Bedrooms | 0.036 | | |
| | (−0.021, 0.092) | | |
| | p = 0.217 | | |
| Number of Beds | 0.041 | 0.052** | 0.042* |
| | (−0.009, 0.090) | (0.006, 0.099) | (−0.007, 0.091) |
| | p = 0.109 | p = 0.029 | p = 0.093 |
| Minimum Nights | −0.002 | | |
| | (−0.005, 0.001) | | |
| | p = 0.139 | | |
| Maximum Nights | 0.0001** | 0.0001* | 0.0001** |
| | (0.00000, 0.0002) | (−0.00000, 0.0002) | (0.00001, 0.0002) |
| | p = 0.048 | p = 0.058 | p = 0.025 |
| Host Acceptance Rate | −0.002* | | |
| | (−0.004, 0.0002) | | |
| | p = 0.071 | | |
| Review Scores Rating | 0.137** | 0.089* | 0.059 |
| | (0.021, 0.253) | (−0.014, 0.193) | (−0.048, 0.167) |
| | p = 0.022 | p = 0.093 | p = 0.280 |
| Host is Superhost (Reference = False) | 0.115*** | 0.114*** | 0.133*** |
| | (0.037, 0.193) | (0.040, 0.188) | (0.055, 0.211) |
| | p = 0.005 | p = 0.003 | p = 0.001 |
| Observations | 364 | 371 | 341 |
| $R^2$ | 0.644 | 0.636 | 0.644 |
| Adjusted $R^2$ | 0.631 | 0.627 | 0.634 |
| Residual Std. Error | 0.317 (df = 350) | 0.317 (df = 361) | 0.319 (df = 331) |
| F Statistic | 48.650*** (df = 13; 350) | 70.023*** (df = 9; 361) | 66.507*** (df = 9; 331) |
| *Note:* | | | *p<0.1; **p<0.05; ***p<0.01 |

## Appendix B

First, an indicator variable that takes the value 1 if price is missing, and 0 otherwise was created. Binary variables were given similar indicator variables. Room type was excluded for being a categorical variable. After these indicators were created, the difference in means for a given variable was calculated, conditional on the presence of price. This calculation is repeated across a large number of bootstrap samples. This gave a bootstrap distribution of a difference in means for each variable considered between listings with missing prices and included prices. Once the distributions are obtained, 95% confidence intervals for this statistic are constructed. If the interval includes 0, the difference is not significant. Otherwise, it is. The bootstrap statistic distributions were normal except for the minimum number of nights and host rating. As such, the intervals for these variables were built using a percentile method, in which the lower bound is the 2.5th percentile and the upper bound is the 97.5th percentile.